

200

Backend Developer Interview Questions with Answers

Master Backend Concepts &
Crack Top Tech Interviews 🚀🔥

**Real Interview
Questions + Answers**

**APIs, Databases,
System Design**

**Beginner to
Advanced Coverage**

**Industry-Level
Preparation**

Perfect for Aspiring Backend Developers 🖥️🔥

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1–30 (Core Backend Basics)

1. What is backend development?
Ans: Backend development involves building server-side logic, databases, APIs, and application architecture.
2. What is a server?
Ans: A system that provides services or resources to other computers (clients).
3. What is an API?
Ans: Application Programming Interface that allows communication between systems.
4. What is REST API?
Ans: An API based on HTTP methods following REST principles.
5. What are HTTP methods?
Ans: GET, POST, PUT, DELETE, PATCH used to perform operations.
6. What is HTTP?
Ans: Protocol for communication between client and server.
7. What is HTTPS?
Ans: Secure version of HTTP using encryption.
8. What is JSON?
Ans: Lightweight data format used for data exchange.
9. What is XML?
Ans: Markup language for structured data.
10. What is CRUD?
Ans: Create, Read, Update, Delete operations.
11. What is a database?
Ans: Organized collection of data.
12. What is SQL?
Ans: Language used to interact with relational databases.
13. What is NoSQL?
Ans: Non-relational database for flexible data storage.
14. What is middleware?
Ans: Software that processes requests before reaching the main logic.
15. What is routing?
Ans: Mapping URLs to specific backend functions.
16. What is MVC architecture?
Ans: Model-View-Controller design pattern.
17. What is authentication?
Ans: Verifying user identity.
18. What is authorization?
Ans: Granting access permissions.
19. What is session?
Ans: Server-side storage of user data.
20. What is cookie?
Ans: Client-side storage data.
21. What is caching?
Ans: Storing frequently accessed data for faster access.
22. What is load balancing?
Ans: Distributing traffic across servers.
23. What is scalability?
Ans: Ability to handle increased load.
24. What is latency?
Ans: Delay in response time.

25. What is throughput?
Ans: Number of requests processed.
26. What is concurrency?
Ans: Handling multiple tasks simultaneously.
27. What is multithreading?
Ans: Multiple threads in execution.
28. What is asynchronous programming?
Ans: Non-blocking execution.
29. What is synchronous programming?
Ans: Blocking execution.
30. What is environment variable?
Ans: Configuration value stored outside code.

31–70 (Databases)

31. What is primary key?
Ans: Unique identifier in a table.
32. What is foreign key?
Ans: Links two tables.
33. What is normalization?
Ans: Reducing redundancy in database.
34. What is denormalization?
Ans: Adding redundancy for performance.
35. What is indexing?
Ans: Improving query speed.
36. What is join?
Ans: Combining data from tables.
37. What is inner join?
Ans: Returns matching records.
38. What is left join?
Ans: Returns all left table records.
39. What is ACID properties?
Ans: Atomicity, Consistency, Isolation, Durability.
40. What is transaction?
Ans: A set of database operations.
41. What is deadlock?
Ans: Two processes waiting on each other.
42. What is replication?
Ans: Copying data across servers.
43. What is sharding?
Ans: Splitting database into parts.
44. What is ORM?
Ans: Object Relational Mapping.
45. What is query optimization?
Ans: Improving query performance.
46. What is stored procedure?
Ans: Predefined SQL code.
47. What is trigger?
Ans: Executes automatically on events.
48. What is view?
Ans: Virtual table.

49. What is schema?
Ans: Database structure.
50. What is migration?
Ans: Updating database structure.

71–120 (APIs & System Design)

71. What is RESTful API?
Ans: API following REST rules.
72. What is stateless API?
Ans: Server does not store client state.
73. What is idempotent API?
Ans: Same request produces same result.
74. What is GraphQL?
Ans: Query language for APIs.
75. What is WebSocket?
Ans: Full-duplex communication protocol.
76. What is rate limiting?
Ans: Limiting API usage.
77. What is API gateway?
Ans: Entry point for APIs.
78. What is microservices architecture?
Ans: Breaking application into services.
79. What is monolithic architecture?
Ans: Single large application.
80. What is service discovery?
Ans: Finding services dynamically.
81. What is containerization?
Ans: Packaging app with dependencies.
82. What is Docker?
Ans: Tool for containerization.
83. What is Kubernetes?
Ans: Container orchestration platform.
84. What is CI/CD?
Ans: Continuous integration/deployment.
85. What is logging?
Ans: Recording system activity.
86. What is monitoring?
Ans: Tracking performance.
87. What is circuit breaker?
Ans: Prevents system failure.
88. What is failover?
Ans: Backup system activation.
89. What is CDN?
Ans: Content delivery network.
90. What is reverse proxy?
Ans: Server that forwards requests.

121–160 (Security)

121. What is encryption?
Ans: Securing data.
122. What is hashing?
Ans: One-way encryption.
123. What is JWT?
Ans: JSON Web Token for authentication.
124. What is OAuth?
Ans: Authorization framework.
125. What is CSRF?
Ans: Cross-site request forgery attack.
126. What is XSS?
Ans: Cross-site scripting attack.
127. What is SQL injection?
Ans: Malicious SQL execution.
128. What is HTTPS handshake?
Ans: Secure connection setup.
129. What is firewall?
Ans: Network security system.
130. What is API security?
Ans: Protecting APIs.

161–200 (Advanced Concepts)

161. What is message queue?
Ans: Asynchronous communication system.
162. What is Kafka?
Ans: Distributed messaging system.
163. What is RabbitMQ?
Ans: Message broker.
164. What is event-driven architecture?
Ans: System reacts to events.
165. What is pub-sub model?
Ans: Publisher-subscriber messaging.
166. What is eventual consistency?
Ans: Data becomes consistent over time.
167. What is CAP theorem?
Ans: Consistency, Availability, Partition tolerance.
168. What is distributed system?
Ans: Multiple systems working together.
169. What is horizontal scaling?
Ans: Adding more servers.
170. What is vertical scaling?
Ans: Increasing server power.
171. What is blue-green deployment?
Ans: Zero downtime deployment.
172. What is canary deployment?
Ans: Gradual rollout.

173. What is serverless architecture?
Ans: No server management.
174. What is Lambda function?
Ans: Serverless execution.
175. What is cold start?
Ans: Delay in serverless startup.
176. What is memory leak?
Ans: Unreleased memory.
177. What is thread pool?
Ans: Collection of threads.
178. What is connection pooling?
Ans: Reusing DB connections.
179. What is API versioning?
Ans: Managing API updates.
180. What is schema evolution?
Ans: Updating database schema.
181. What is load testing?
Ans: Testing system under load.
182. What is stress testing?
Ans: Testing beyond limits.
183. What is unit testing?
Ans: Testing individual components.
184. What is integration testing?
Ans: Testing combined modules.
185. What is mocking?
Ans: Simulating dependencies.
186. What is dependency injection?
Ans: Providing dependencies externally.
187. What is clean architecture?
Ans: Separation of concerns.
188. What is SOLID principles?
Ans: Object-oriented design rules.
189. What is design pattern?
Ans: Reusable solution.
190. What is singleton pattern?
Ans: Single instance class.
191. What is factory pattern?
Ans: Object creation pattern.
192. What is repository pattern?
Ans: Data access abstraction.
193. What is CQRS?
Ans: Separate read/write operations.
194. What is event sourcing?
Ans: Storing events instead of state.
195. What is observability?
Ans: Understanding system state.
196. What is tracing?
Ans: Tracking request flow.
197. What is SLA?
Ans: Service agreement.
198. What is uptime?
Ans: System availability.

199. What is rollback?

Ans: Reverting changes.

200. What is DevOps?

Ans: Development + Operations practices.

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